

# Fragmented Accountability and Under-explained Human Oversight in the United States Federal AI Inventory

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Federal Artificial Intelligence (AI) inventories are increasingly framed as transparency mechanisms for documenting how public agencies adopt, classify, and govern algorithmic systems. In this paper, we analyze the 2025 United States Federal Agency AI Use Case Inventory as a public-facing governance artifact and disclosure infrastructure, combining quantitative mapping of 3,611 individually reported use cases across 56 agencies with qualitative analysis guided by the Algorithmic Decision-Making Adapted for the Public Sector framework. Our findings show that the inventory renders federal AI unevenly visible: reported use cases are concentrated across agencies, procurement and governance-related details are frequently incomplete, and impact classifications and claims of human oversight are often under-explained. We discuss how the inventory functions as an imbricated governance infrastructure, how classification performs boundary work in making AI systems governable, and how accounts of human-AI collaboration disclose formal oversight while often obscuring situated practices of judgment, contestability, and responsibility.

## 1 Introduction

Artificial Intelligence (AI) inventories are becoming increasingly common mechanisms for public-sector transparency, accountability, and oversight in governments worldwide [69]. In the United States (US), federal AI inventories are grounded in reporting requirements established through Executive Order-13960 [75], Advancing American AI Act [78], and Office of Management and Budget (OMB) Memorandum-M-25-21 [72]. These policies require federal agencies, with limited exceptions<sup>1</sup>, to conduct annual inventories of AI use cases and make publicly releasable information available in machine-readable form [73, 74]. The resulting 2025 US Federal Agency AI Inventory offers a large-scale view of federal AI use across agencies. This paper examines the inventory as a public-facing governance artifact that makes federal AI activity administratively legible while also revealing how claims of human-AI complementarity, oversight, and alignment are organized.

Public-sector AI governance requires abstract principles such as transparency, fairness, accountability, privacy, and trustworthiness to be implemented through bureaucratic organizations with different mandates, procurement arrangements, and technical expertise [36, 95, 108]. Documentation mechanisms such as audits, impact assessments, registers, and inventories can make parts of these organizational arrangements inspectable [79, 82, 109]. However, they should not be interpreted as straightforward transparency tools, especially in cross-agency contexts where visibility is shaped by different reporting practices, institutional priorities, and forms of technical and administrative capacity [2, 51, 80]. Here, our central insight is that the federal AI inventory should not be read only as a dataset of AI systems, but as a disclosure infrastructure of bureaucratic governance. In this paper, we understand infrastructures as sociotechnical arrangements that coordinate actors, categories, routines, standards, and institutional practices across organizational boundaries [11, 102]. The inventory not only reports federal AI activity but also organizes how AI becomes visible, comparable, and governable. Hence, its structure, omissions, and justifications are analytically meaningful. We ask: *How does the US Federal AI Use Case Inventory render public-sector AI governable, and where does this visibility fall short for accountability and contestability?*

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<sup>1</sup>Excluded cases include certain national security, intelligence, research, defense-related, and legally restricted uses [73].

Empirically, we analyze 3,611 individually reported AI use cases from 56 federal agencies, along with consolidated commercial off-the-shelf (COTS) AI reporting materials [74]. We combine descriptive quantitative analysis with theoretically guided critical discourse analysis. Rather than treating the inventory as a complete account of federal AI activity, we use the Algorithmic Decision-Making Adapted for the Public Sector (ADMAPS) framework [91] to examine how it is shaped by the interplay among bureaucratic processes, algorithmic decision-making, and human discretion. We find that while the inventory substantially expands public visibility into federal AI use, that visibility remains uneven and often insufficient for accountability. Our study examines how the missing fields, uneven reporting, ambiguous classifications, and repeated invocations of human review are not merely data quality problems, but are evidence of how agencies interpret AI governance requirements and translate them into public documentation. We make three contributions. First, it shows that AI use cases, especially the high-impact ones, are concentrated among a small number of agencies and explains how impact determinations are shaped by technical classifications, policy domains, reporting practices, and interpretive judgment. Second, it explains how the inventory standardizes AI reporting while exposing fragmentation through missing fields, ambiguous classifications, and incomplete procurement information. Third, it argues that the inventory produces *visibility without full contestability*: agencies often invoke human oversight and recommendation-only functionality to justify lower-impact classifications, but rarely explain how officials evaluate, weigh, or contest AI outputs. We conclude by reflecting on these findings to identify concerns, challenges, and design recommendations for human-AI collaboration in public-sector governance.

## 2 Literature Review

### 2.1 Public-Sector AI Governance as Bureaucratic Implementation

Public-sector AI governance is often framed through principles such as transparency, fairness, accountability, safety, privacy, and trustworthiness [28, 34, 46]. However, these principles must be implemented by bureaucratic organizations with different missions, legal obligations, resources, procurement arrangements, and technical capacities [36]. Prior work shows that algorithmic accountability is not only a question of technical system design, but also of institutional context, professional judgment, organizational incentives, and legal mandates [47, 95, 108]. These challenges are amplified in the public sector because algorithmic systems may shape access to welfare, housing, child protection, fraud detection, and other public services [22, 30, 84, 85]. Such systems can make state decisions harder for affected communities to understand, challenge, or appeal [13, 83].

In the US context, Lawrence et al. showed that federal agencies have struggled to implement binding AI governance mandates, including public AI use case inventories, due to policy ambiguity, uneven capacity, staffing constraints, limited technical expertise, and inconsistent reporting practices [56]. This highlights that transparency mandates do not implement themselves; agencies must identify relevant systems, interpret reporting requirements, coordinate across internal offices, document procurement relationships, and sustain public records over time. Recent work on US state AI policy similarly shows that American AI governance is fragmented across committees, reports, executive actions, and state-level initiatives, with policy discourse often emphasizing benefits more than risks and omitting broader sociotechnical concerns [65].

Public-sector AI also complicates accountability because algorithmic systems often redistribute rather than eliminate discretion [1, 15, 91]. Public officials exercise judgment when translating rules into practice [59]. Through empirical studies across domains, such as homeless shelters [67], immigration [20], child welfare [91, 93], computer-supported cooperative work and public-sector AI research similarly show that algorithmic decision-support systems shift responsibility among frontline workers, managers, vendors, agencies, and technical systems [41, 108]. Hence, claims that

AI systems are merely advisory, recommendation-only, or subject to human review should be carefully scrutinized. Human oversight is not a self-explanatory safeguard [40]. It is an organizational practice shaped by training, workload, expertise, institutional values, and the procedures through which officials evaluate algorithmic outputs [104].

## 2.2 Inventories and Registers as Disclosure Infrastructures

AI inventories and registers belong to a broader family of documentation and disclosure mechanisms intended to make algorithmic systems inspectable [20, 79]. However, they are not merely descriptive databases but disclosure infrastructures—sociotechnical arrangements that coordinate information and standardization among various actors and institutions. These infrastructures—both process and outcome—vary significantly across geopolitical contexts, objectives, audiences, and organizational structures [79]. Their organizational routines are learned as part of membership in communities of practice, built on standards and classifications, and often become most visible when they break down or require repair [11, 102, 103]. This insight highlights the hidden human labor [57], such as identifying systems, interpreting reporting rules and translating contextual practices into standardized fields, that underlie such disclosure infrastructure or documentation.

Documentation frameworks such as datasheets and model cards standardize information about data provenance, intended use, limitations, risks, and performance [38, 66]. Audit reports [76, 82], impact assessments [20, 109], model documentation [111], data statements [62], and reporting standards [100] similarly translate abstract accountability commitments into artifacts that can circulate among developers, regulators, organizations, and publics. However, documentation does not simply describe technical systems. It shapes which properties become visible [80], which actors are rendered responsible [54], and what kinds of accountability claims [10, 81] can be made.

Critical data documentation scholarship shows that standardized artifacts can both enable and constrain accountability. Claims about scale, representativeness, neutrality, or completeness may obscure the social and organizational conditions through which data are produced and maintained [24, 50]. Similarly, documentation practices can reduce fairness and accountability concerns to questions of data sufficiency while rendering data labor, annotation work, and institutional context less visible [3, 45, 94, 96]. Work on large-scale data infrastructures such as Common Crawl [4] and critiques of “lazy” data documentation [99] further shows how disclosure can diffuse responsibility across actors, licenses, downstream users, and methodological shortcuts. These insights are useful for analyzing AI inventories as they highlight how disclosure artifacts like registers and inventories define what kinds of information count as relevant for public accountability.

Algorithm registers and AI inventories extend documentation from datasets and models to public institutions. Municipal governments such as Amsterdam [68], Helsinki [43], Nantes, Antibes, and Lyon [77] were early adopters of algorithm registers, publishing structured information about algorithmic systems used in public services [98]. Similar disclosure mechanisms have emerged or been proposed across jurisdictions, including the European Union’s Digital Services Act [52], Bangladesh’s AI policy [71], Canadian federal AI registers, and American AI inventories [56, 74]. Empirical studies of transparency databases [48], AI registers [69], and ethical charters [39] show that templates, reporting thresholds, machine-readable fields, and organizational incentives shape what institutions disclose and what remains difficult to interpret [29, 44, 105]. These systems may improve public visibility while still falling short of regulatory goals, especially when they provide high-level descriptions without sufficient contextual explanation or enforceable accountability [18, 36, 42].

In a recent work, Pi et al. argue that algorithm registers vary substantially across jurisdictions and can function as mechanisms for public disclosure, regulatory coordination, pre-market approval, ecosystem-level monitoring, and broader governance infrastructure depending on their design and

institutional context [79]. Similarly, examining the Federal Government of Canada's AI Register, a recent work [21] explained why these registers are not just neutral mirrors of government activity but artifacts that configure the boundaries of accountability by revealing some aspects of AI systems while obscuring bureaucratic practice, uncertainty management, and human discretion. These studies emphasize that the AI inventories, such as the one published by the US federal government, should also be analyzed from a sociotechnical perspective to understand how they simultaneously shape and are shaped by administrative practice.

### 2.3 Classification and the Boundaries of Accountable Disclosure

AI inventories are both outcomes and parts of broader sociotechnical arrangements, i.e., infrastructures that coordinate actors, practices, and institutions across organizational boundaries [103]. Their "ontological design" [20] organizes the boundaries of accountability of public sector AI. Accountability is commonly understood as the capacity to trace, justify, and assign responsibility for algorithmic decisions and outcomes [8, 26, 110], while transparency refers to the availability of information that enables explanation, documentation, or disclosure of system properties [6, 55, 107]. However, transparency alone does not guarantee accountability, public understanding, or contestability [2, 51].

This distinction is particularly important where affected people may need to understand, challenge, or seek redress for AI-mediated decisions [90]. In governance contexts, transparency is better understood as a situated communicative arrangement or "disclosure by design," in which information is audience-specific, shaped by institutional incentives, and structured by power relations [31, 70]. Algorithmic supply chains can fragment responsibility across agencies, contractors, developers, and vendors, making traceability central to accountability [17, 53]. Algorithmic audits can surface bias, error, or harm [82, 87]. Both researchers and everyday users have used algorithmic audits to scrutinize AI systems [23, 63, 97], like recommendation and search algorithms [5, 86], computer vision (CV)-based processes (e.g., generative art [101], image captioning [112], facial recognition [14]), and natural language processing (NLP) technologies (e.g., sentiment analysis [19, 49], hate-speech detector [88], machine translation [89], text generation [33]). However, output-focused audits are often insufficient [37], especially when accountability questions concern procurement, training data, privacy review, evaluation practices, human oversight, and organizational workflows [16, 60, 92]. In such cases, investigative and journalistic approaches, combining technical inspection with document analysis, interviews, and policy tracing, are also effective in identifying opaque practices and institutional accountability gaps [21, 25, 27].

Prior work by Das and colleagues [21] used the Algorithmic Decision-Making Adapted for the Public Sector (ADMAPS) framework [91]. It conceptualizes public-sector algorithmic decision-making as an interaction among bureaucratic processes, algorithmic decision-making, and human discretion. Bureaucratic processes include resources, constraints, administrative routines, procurement, training, and governing laws and policies. Algorithmic decision-making includes data, decision-support functions, uncertainty, and accountability mechanisms. Human discretion includes professional expertise, value judgments, and interpretive practices. We used the ADMAPS framework to examine the US Federal AI Use Case Inventory from a sociotechnical perspective. Because the inventory explains how and why federal agencies procure, categorize, and use AI systems, ADMAPS provides a useful lens for examining it as a governance artifact.

3 Methods

3.1 Data Collection and Overview

Our study draws on the 2025 US Federal Agency AI Use Case Inventory, a public repository maintained by OMB that consolidates agency-level AI inventories across the federal government [74]. The repository includes both individually reported and COTS AI use cases. At the time of our study, it contained submissions from 56 agencies, including 3,611 individually reported AI use cases. OMB also provides supporting materials, such as data dictionaries, sourcing summaries, and standardization reports, documenting how heterogeneous agency submissions were harmonized.

We treat the inventory not as a complete census of federal AI activity, but as a standardized disclosure artifact shaped by reporting requirements, transparency obligations, and agency-specific interpretations of OMB guidance. This is particularly important because the 2025 reporting process reduced the number of reporting questions and introduced consolidated reporting for common COTS uses [74], influencing what becomes visible as a reportable AI use case and what remains consolidated, withheld, or outside the inventory’s public scope.

3.2 Data Analysis

We first conducted exploratory quantitative analysis using descriptive statistics and visualizations to identify patterns of concentration, variation, and missingness across agencies and governance fields. We then conducted a qualitative analysis of the inventory’s textual fields. Using the ADMAPS framework [91] as the analytical lens, the first author conducted initial coding, identifying recurring representational patterns, such as actors, tasks, objectives, benefits, and risks. Drawing on critical discourse analysis (CDA), our coding treated both what is emphasized and what is left unsaid in texts and documents as analytically meaningful [32, 64]. Using each AI use case entry in the inventory as the unit of analysis, he iteratively developed a set of codes. The generated codes were refined through discussions with two supervising authors, during which disagreements or ambiguities were documented in analytic memos and resolved collaboratively. Codes were then merged, split, or adjusted to maintain consistency. Following social computing research guidelines for such approaches [61], we did not calculate inter-coder reliability scores. To examine how federal agencies frame and govern AI systems, we mapped the resulting codes to the three ADMAPS dimensions [91]. For example, codes that explicitly reference AI classifications, model inputs, and data types were mapped to the algorithmic decision-making dimension. Throughout the qualitative analysis, we iteratively juxtaposed the emerging codes with corresponding quantitative patterns.

3.3 Limitations

Our analysis is limited to what is disclosed in the inventory and cannot capture undocumented systems, informal practices, or the use of AI in everyday bureaucratic work. Therefore, we examine the politics of disclosure and representation rather than the operational performance or impacts of individual systems. Again, while our analysis is interpretive and guided by ADMAPS [91], alternative frameworks might emphasize different aspects of the inventory.

4 Results

We analyze the US federal AI inventory as a socio-technical governance infrastructure that documents interactions among bureaucratic processes, algorithmic decision-making, and human discretion. Our ADMAPS-guided analysis shows how different agencies and departments frame expertise, manage administrative constraints, classify impact, and document accountability. However, these dimensions are not uniformly visible. Some aspects of AI governance are well-documented, whereas others remain fragmented, sparsely quantified, inconsistently reported, or largely absent.

## 4.1 Bureaucratic Processes

The inventory provides insight into AI development, adoption, and governance through agency reporting patterns, lifecycle stages, procurement information, and impact classifications. Their substantial variation demonstrates that federal AI governance is unevenly enacted across agencies, reflecting diverse institutional contexts, capacities, and interpretations of policy requirements. These bureaucratic processes are infrastructural because they shape how AI systems are identified, categorized, procured, monitored, and made publicly reportable.

**4.1.1 Administrative Scale and Resource Constraints.** Agencies reported challenges, including high volumes of public submissions, limited staffing capacity, extensive document collections, cybersecurity workloads, and labor-intensive administrative processes. For example, the Securities and Exchange Commission (SEC)-37 noted that it receives “*high volumes of Tips, Complaints, and Referrals*” while having “*limited staff to triage*” them. Other agencies (e.g., National Aeronautics and Space Administration (NASA), National Science Foundation, SEC, Department of Housing and Urban Development) described manual review processes as “*time-consuming*,” information retrieval as burdensome, and data integration as difficult. Hence, AI adoption often appears to be a response to bureaucratic scarcity. However, AI use cases were unevenly distributed across agencies. Eight agencies accounted for approximately 72% of the inventory (see Figure 1). The agencies with the largest number of reported use cases were the Department of Health and Human Services (HHS) (447), NASA (425), Department of Veterans Affairs (VA) (367), Department of Energy (DOE) (340), Department of Justice (DOJ) (314), Department of the Interior (DOI) (247), Department of Homeland Security (DHS) (238), and Department of Commerce (DOC) (223). In contrast, many agencies reported only a few.

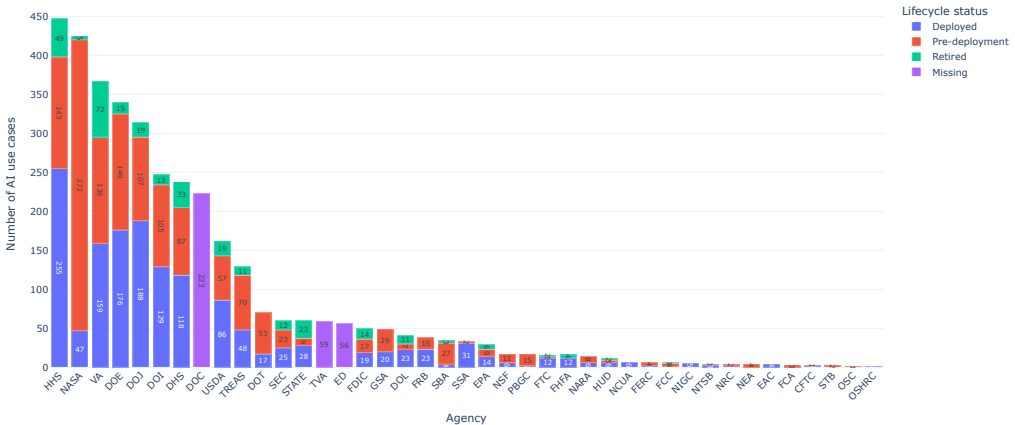


Fig. 1. Number of individually reported AI use cases by agency. Agency acronyms follow the federal agency abbreviations listed by USA.gov’s index of government departments and agencies [106].

Again, among the reported use cases in the inventory, 41% were marked as pre-deployment. Other systems were reported as deployed (28.8%), pilot (12.2%), or retired (8.7%). Lifecycle status was missing for 338 entries—in some cases for entire agencies (e.g., DOC; see Figure 1). This mixed



lifecycle composition indicates that the inventory captures prospective and emerging AI activity, not only systems already authorized. The presence of retired systems shows that the inventory also preserves traces of discontinued AI activity. Overall, this apparent temporal comprehensiveness is shaped by the reporting practices of a relatively small set of agencies and by a mix of systems that are being imagined, acquired, tested, deployed, and retired.

**4.1.2 Administration, Procurement, and Organizational Capacity.** Most entries (55.69%) do not report how the corresponding AI systems were procured. Among others, 17.92% were purchased from a vendor, 16.58% were developed in-house, and 9.8% were developed with both. With 453 unique contractors, federal agencies do not appear to depend heavily on any single vendor for specific categories of AI systems, such as classical machine learning (ML) or CV. In most cases, no vendor supplied more than ten use cases within a given AI classification (see Figure 2). Generative AI (GenAI) was a notable exception, where federal agencies relied disproportionately on a small set of vendors, particularly Microsoft (70 use cases), OpenAI (17), Google (16), and Palantir (12), suggesting an emerging concentration of procurement.

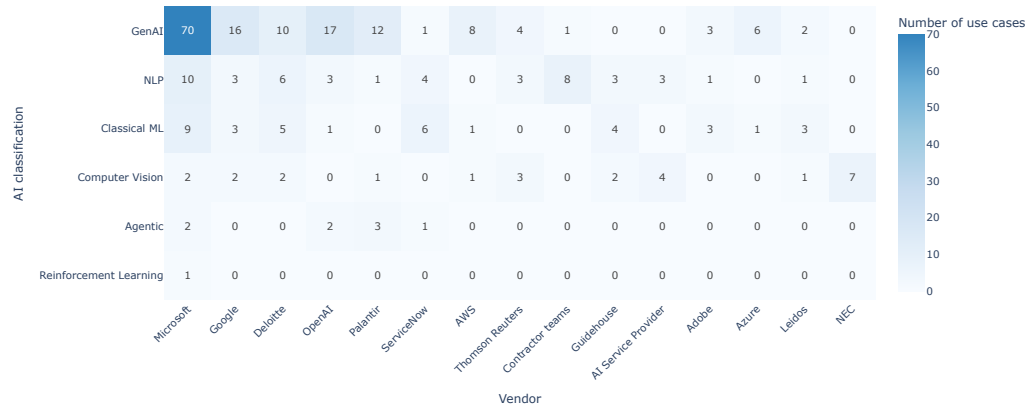


Fig. 2. Vendor (top 15) dependence by AI classification.

Specific COTS product names, license information, and user estimates were missing in almost half of the entries. Even when product names were provided, naming conventions varied considerably. Moreover, organizational capacity itself was often poorly documented. Fields related to operator training, monitoring, and governance practices were overwhelmingly incomplete. Only 4.09% of use cases reported established operator training, while 90.68% left the field blank. Similarly, only 2.41% reported ongoing monitoring by an independent reviewer, while 91.19% provided no information.

**4.1.3 Law and Policy Through High-Impact Classification.** Fragmentation is especially visible in the distribution of impact classifications. High-impact systems were not distributed evenly across agencies. VA alone reported 215 high-impact systems, accounting for nearly half of all high-impact entries. DOJ reported 114 high-impact systems, DHS reported 55, and DOE reported 29. Together, these four agencies accounted for 413 of the 431 high-impact systems (approximately 96%) in the

inventory. This concentration contrasts sharply with agencies that reported many AI use cases but very few high-impact systems. Among the 425 use cases reported by NASA, only one was a high-impact system. DOI reported 247 use cases, all classified as not high-impact. These contrasts suggest that the high-impact classification rate is not simply proportional to the number of AI use cases an agency reports. The rate of high-impact classification further illustrates this point. Among agencies with more than 5 reported use cases, four agencies: VA (52%), DOJ (28%), DHS (13%), and DOE (7%), account for almost all high-impact cases. However, impact status was missing for 9% entries, and these were not evenly distributed across agencies. For example, DOC, despite reporting one of the largest numbers of AI use cases, did not provide impact status for any of them. These differences show that high-impact classification is mediated by agency-specific domains of action and agency-specific judgments about impact, risk interpretation, and reporting practices.

High-impact classification also varied substantially across topic areas. Some domains were much more likely to include high-impact systems than others. Health and Medical Processing had 139 high-impact entries out of 181 total entries, or approximately 77%. Government Benefits Processing had 20 high-impact entries out of 37, or approximately 54%. Law Enforcement had 130 high-impact entries out of 280, or approximately 46%, along with 57 entries that were presumed high-impact but determined not high-impact. Other topic areas had much lower high-impact rates. Only approximately 1.5% entries were high-impact for Science, the largest topic area. Information Technology (IT) included 11 high-impact entries, accounting for approximately 4% of use cases in this area. Administrative Functions included 17 high-impact entries out of 289, or approximately 6%. These differences suggest that AI systems used in domains that shape health-related decisions, access to public benefits, legal practices, or other forms of public consequence are much more likely to be classified as high-impact than those in science, internal administration, or IT support.

## 4.2 Algorithmic Decision-Making

The inventory offers relatively detailed visibility into some of these technical dimensions of through classifications, topic areas, and descriptions of intended outputs. However, details about certain background aspects of the federal AI infrastructure, such as training data, demographic variables, and contestability pathways, are often incomplete.

*4.2.1 Technical Classification and Forms of Decision Support.* Classical or predictive ML was the largest category in the inventory (1,114 entries). It was followed by GenAI (858), NLP (454), CV (292), Agentic AI (117), and reinforcement learning (RL) (12). However, 662 entries did not report any AI classification. While OMB's classification scheme distinguishes among the aforementioned AI system classifications, federal agencies often describe their use cases through a smaller set of recurring decision-support functions. For example, SEC-23 described how it sought to help staff "search and analyze a corpus of unstructured filings" using NLP. Overall, we identified four dominant forms of decision support across the inventory: prediction, classification, retrieval/summarization, and recommendation. Systems generated outputs ranging from anomaly scores, risk indicators, and classifications to dashboards, legal research assistance, summaries, and retrieval-augmented responses. Even highly technical systems frequently produced information intended to guide human action rather than automate decisions directly. The consolidated COTS inventory reaffirms this pattern in procurement and commercial AI use: the most commonly affirmed uses included generating first drafts of documents or briefing materials, using AI-assisted tools in word processors, summarizing lengthy reports, searching agency information, transcribing or summarizing virtual meetings, improving written communication, and generating code.

Across AI classifications, CV had the highest high-impact rate, with 116 high-impact entries out of 267, or approximately 43.5%. High-impact classification rates for other AI classifications



were as follows: Agentic AI (25.6%), NLP (18%), Classical ML (12.4%), and GenAI (8.6%), with all RL categorized as not high-impact. These distributions suggest that a use case’s impact classification is not reducible to its technical category alone. Classical and emerging AI systems can appear in low- and high-impact contexts, depending on the domain of use and the consequences.

**4.2.2 Relevant Data and Partial Visibility Into Model Inputs.** In the federal inventory, several fields could support accountability related to the materials and technical processes through which AI systems are built and evaluated. However, in practice, these fields were weakly documented. For example, most of the inventory did not describe what kinds of data were used to train, fine-tune, or evaluate models (64%); whether they involved personally identifiable information (PII) (58.9%); or whether they used any demographic data (69.4%). Only 10.84% of entries explicitly report using PII. While 27.6% reported using no demographic data, fewer than 1% entries mention what demographic variables are used (see Figure 3 for their breakdown).

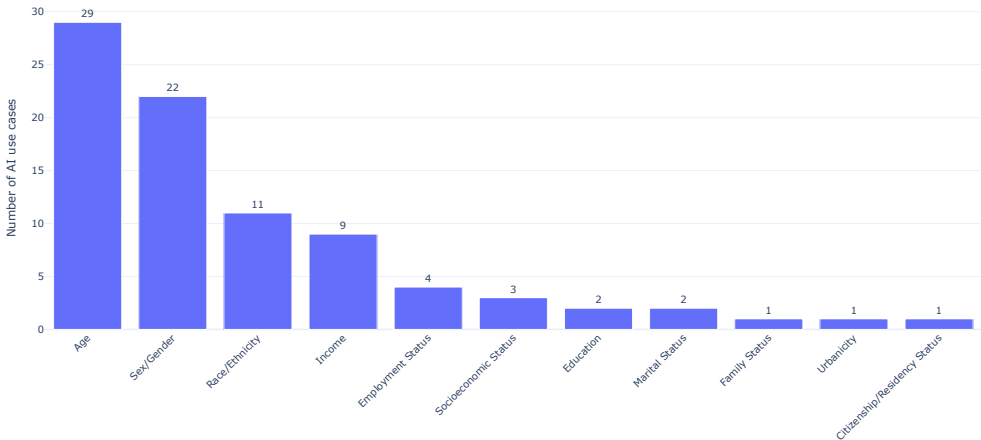


Fig. 3. Demographic Variables Explicitly Used as Model Features

Privacy Impact Assessment (PIA) links were available for only 3.3% entries. While 17.5% use cases include custom-developed code, only a few of those are open source. These patterns limit the inventory’s usefulness for external technical inspection. The inventory can identify discrete aspects of AI systems (e.g., whether they involve PII, use custom-developed code, or have PIAs), but most entries do not provide sufficient information to comprehensively assess data provenance, privacy review, model evaluation, or implementation transparency.

Thus, the inventory provides only limited visibility into the data foundations of federal AI systems. Most entries do not disclose training data, demographic variables, evaluation data, or privacy review documentation. Although some systems explicitly mention demographic attributes such as age, race, sex, education, and family structure, fewer than one percent of entries specify demographic variables in sufficient detail to support external scrutiny.

**4.2.3 Degrees of Uncertainty, Contestability, and Accountability.** Compared to most AI use cases in the inventory (63.74%) that were reported as eligible for public disclosure, only 0.69% were marked

for withholding based on agency determination, and 0.14% due to legal prohibitions. However, this reporting field was left unanswered for 1,027 entries (35.33% of all records). Because the relevant governance fields were overwhelmingly incomplete, it was often difficult to assess the transparency of federal AI systems or hold them accountable. Only 0.72% of use cases reported an established appeal process, and 4.06% are in the process of establishing one, while 90.40% left the field blank. Similarly, 90.37% did not report fail-safe mechanisms, 92.29% did not report whether an AI impact assessment had been completed, and 91.50% did not report whether pre-deployment testing had been conducted. Though most AI use cases (56.66%) did not report whether they had an associated Authorization to Operate (ATO), an almost equal number reported having and not having one. While this complicates the security, privacy, and impact landscape for AI use across various agencies, the entries with ATOs often do not outline the scope of those authorizations.

Many entries provided only procedural descriptions such as “*consistent with Executive Orders and OMB guidance*” (DOJ-0324, DOJ-0272), offering little substantive discussion of uncertainty, risk, bias, privacy harms, or downstream consequences. Even when impacts were discussed, they were often framed in general terms such as efficiency gains, improved security, or possible misidentification risks. Consequently, the inventory provides substantial visibility into the existence and intended purpose of AI systems while offering comparatively limited visibility into how uncertainty is evaluated, monitored, contested, or governed in practice.

### 4.3 Human Discretion

Across the inventory, agencies consistently framed AI as an augmentation rather than a replacement for bureaucratic judgment. The open-ended descriptions provide insight into how agencies frame expertise, articulate the benefits of AI adoption, and justify their impact classifications by referencing human oversight and review.

**4.3.1 AI as Support for Professional Expertise.** Many agencies sought to address a recurring concern with information overload and the practical limits of professional expertise. Agencies described difficulties reviewing large document collections, navigating complex regulatory materials, analyzing unstructured information, and conducting time-intensive research. For example, SEC-25 aimed to “*help staff more efficiently search and analyze disclosure text contained in various filings.*” Similar motivations appeared in legal research applications, internal knowledge bases, and cybersecurity operations. The expected benefits and outputs of these systems further illustrate their role as expertise-support tools. Agencies reported outputs such as document summaries, dashboards, recommendations, reports, metadata generation, visualizations, and conversational responses. For example, SEC-44 generated “*document summaries, document drafts, workflow analysis, and answers to user-selected questions*”, while SEC-50 provided “*real-time answers to staff questions about disclosure rules, filing procedures, and review protocols.*” Rather than replacing staff expertise, these outputs were intended to make expertise more scalable and accessible.

The distribution of AI classifications across different tasks and topic areas reflects the increasing use of GenAI. Figure 4 shows that different forms of AI support different kinds of expertise. Classical ML is predominantly used in scientific and analytical activities across different agencies, whereas GenAI is commonly used in administrative functions (e.g., internal knowledge management, staff support) and information technology (e.g., software development and engineering, IT operations and infrastructure management, data analytics and business intelligence).

**4.3.2 Prioritization of Certain Value Judgments.** Expected benefits descriptions highlight the values that agencies associate with AI adoption. Many agencies explicitly framed AI as a response to limited administrative capacity. For example, agencies described improving “*efficiency by automating*



Fig. 4. AI Classification by Use Case Topic Area

previously manual steps" (SEC-65), enabling "more efficient legal research" (SEC-44), reducing workload associated with document review, improving timeliness of investigations, and helping staff focus on "higher-value activities" as described for DHS-2711. Nuclear Regulatory Commission (NRC) use case, NRC-1, stated that AI would improve the timeliness of "risk- and policy-informed decisions," while another, NRC-2, argued that AI could improve "efficiency, accuracy, and responsiveness."

However, explicit discussions of values such as fairness, participation, and contestability were absent from many of these descriptions. Instead, agencies frequently justified AI in terms of administrative values such as efficiency, speed, consistency, productivity, scalability, risk management, and improved access to information. In this sense, the inventory reveals not only what AI systems do but also the kinds of institutional values that agencies prioritize when describing them.

**4.3.3 Human Discretion as Under-explained Logic.** Across agencies, AI systems were frequently described as recommendation tools, assistive systems, or informational resources whose outputs required human validation. For example, agencies (e.g., HHS, Department of Labor (DOL)) repeatedly justified systems as not high-impact because "humans retain full control," "mandatory human review" was required, or AI outputs were merely "one of several informational inputs." HHS explained that an AI-based Freedom of Information Act redaction system was not high-impact because it "operates as a recommendation system only, with mandatory human review and approval required for all outputs." Another HHS entry stated that AI-generated referral recommendations undergo "100% human review," while a third emphasized that final decisions remained the responsibility of agency personnel. Many of these explanations from DHS and DOL directly echoed the language of OMB Memorandum [72], particularly the criterion that AI outputs must not "serve as a principal basis" for decisions affecting rights, opportunities, or access to services, framing human discretion as the mechanism through which AI-related impacts are kept in check.

However, the inventory provides little insight into how this human oversight is actually exercised in practice. It rarely describes the cognitive, organizational, or procedural processes through which the federal officials evaluate AI-generated recommendations, reconcile them with other sources of information, or arrive at final decisions. The interpretive character of the impact classification

category “*presumed high-impact, but determined not high-impact*” illustrates this. The inventory includes 97 such entries (3.34%), concentrated primarily within DHS (49) and DOJ (32). This category suggests that impact classification is not a purely mechanical exercise based on predefined criteria. Rather, these determinations often relied on arguments about human oversight, advisory, and agency review. However, the inventory offers limited visibility into what evidence, review procedures, or evaluative standards were used to move systems from presumed high-impact to not high-impact status. Therefore, while human discretion is presented as a solution to AI-related risks, it offers far less transparency into how the underlying reasoning leads to the documented annotations.

These descriptions of assistive systems, recommendation-only outputs, and retained human authority constitute the inventory’s primary account of human-AI collaboration, but this collaboration is visible mainly as a formal governance claim rather than as a situated practice.

## 5 Discussion

Federal AI inventory substantially expands public visibility into AI use across government, providing a common reference point for policymakers, oversight bodies, researchers, journalists, civil society organizations, and affected communities. However, it only offers a partial and uneven view of the broader federal AI landscape. A high number of reported use cases may indicate extensive AI adoption, but it may also reflect broader interpretations of what constitutes reportable AI, stronger internal reporting infrastructures, or greater willingness to disclose AI activity. Conversely, a smaller number of reported use cases does not necessarily indicate limited AI use. Here, we reflect on these findings through three related lenses: the inventory as governance infrastructure, classification as boundary work, and human-AI collaboration as materially enacted judgment.

### 5.1 The Inventory as Imbricated Governance Infrastructure

The inventory surfaces the challenges of maintaining a shared governance infrastructure across agencies with different capacities, priorities, procurement arrangements, and interpretations of reporting requirements. Beyond the information it contains, the inventory’s significance lies in the organizational relationships and practices it exposes. The concept of *imbrication* helps explain this process: human and material agencies become interlocked over time, producing and reshaping routines and technologies [58]. The outcome of the imbrication of prior organizational work (e.g., OMB guidance), structured fields, agency-submitted reports, policy interpretations, procurement arrangements, and technical classifications translates situated AI practices across agencies into standardized disclosure fields producing the federal AI inventory. Thus, the inventory becomes a material-discursive infrastructure through which federal AI becomes reportable.

The infrastructural perspective also reframes missingness. Several structured fields and open-ended governance questions are frequently incomplete. Such inconsistency and incompleteness, at times for entire agencies, are not simply noise or data quality issues. They mark points where the disclosure infrastructure depends on unresolved coordination, uneven administrative capacity, or competing interpretations of what should be reported. For example, procurement-related omissions are especially consequential because COTS AI systems entangle the federal government with vendors through licensing arrangements, embedded defaults, and proprietary models that remain only partially visible to the public.

### 5.2 Classification as Boundary Work in Reportable AI

If infrastructure makes federal AI systems visible, classification determines how they are seen. Though classifications appear natural once established, they are the product of negotiations, compromises, and judgments [35]. The inventory relies on multiple classification schemes. The underlying coordination across agencies transforms heterogeneous systems and practices into a

standardized administrative ontology, which defines the boundaries of what becomes governable, comparable, and accountable [12]. However, we found that these boundaries are often under-explained. For example, the distinction between Classical ML and NLP is not always self-evident. If a system uses a support vector machine to analyze a text corpus, should it be classified as Classical ML because of the underlying modeling technique, or as NLP because the application domain involves language data? While such cases require articulation work to reconcile standardized categories with local organizational contexts, the inventory's reliance on closed-ended fields often leaves little room for agencies to explain how such judgments are made.

This limitation is especially important because classification has governance consequences. For example, the category of "presumed high-impact, but determined not high-impact" signals that agencies engage in interpretive work to justify why some systems that initially appear consequential should be exempted from a more stringent classification. However, the inventory rarely discloses the assumptions, evidence, organizational negotiations, or review procedures through which such determinations are made. While OMB documentation provides high-level guidance and the data dictionary helps resolve low-level inconsistencies (e.g., different labels that denote the same classification), a more accountable inventory would include more discursive fields through which agencies explain boundary cases, overlaps, exceptions, and interpretive judgments.

### 5.3 Human-AI Collaboration as Materially Enacted Judgment

Many AI use cases relied on narratives of retaining human authority with human review or recommendation-only functionality to justify why systems were not considered high-impact. However, these descriptions rarely explain how officials evaluate AI-generated recommendations, weigh competing sources of information, or exercise discretion in practice. While discretionary judgment is central to how policies are enacted [59] and often involves complex organizational and socioculturally grounded interpretive work rather than simple review procedures [91], the inventory rarely explains how such judgment is exercised in specific human-AI collaborations.

This creates a central accountability tension, especially because technologies do not simply support or mediate work, but participate in how work is materially enacted through particular artifacts, bodies, spaces, temporal arrangements, and infrastructures [7]. From this perspective, AI-generated summaries, dashboards, risk indicators, and recommendations do not simply provide neutral inputs but reshape what officials notice, which evidence is prioritized, how expertise is coordinated, and how decisions become actionable. Thus, even when agencies describe AI systems as assistive or advisory, those systems may materially reorganize the conditions under which human discretion is exercised. This also complicates the inventory's treatment of human oversight as a safeguard. Human judgment matters in algorithmic systems partly because of its potential to support individual justice: the assessment of each case on its own merits rather than only through comparison with generalized patterns or prior cases [9]. However, the inventory often treats human involvement as a categorical safeguard rather than as a situated practice. It tells readers that humans remain in the loop, but rarely shows whether those humans have the authority, time, expertise, willingness, and institutional support needed to meaningfully review, override, justify, or contest AI outputs. Therefore, like many prior public algorithmic transparency artifacts [2, 21, 51], the US AI inventory falls short in supporting contestability.

## 6 Conclusion

The US Federal AI Inventory renders AI governable through the categories, omissions, and justificatory practices that define what counts as reportable. AI is not simply discovered and disclosed, but is identified and recognized through bureaucratic routines, policy interpretations, procurement arrangements, technical classifications, and claims about human oversight. From this view, the

inventory's gaps are consequential: missing procurement information obscures institutional and commercial dependencies; non-discursive classifications constrain public comprehension; lack of pathways to appeal limits contestability. Accountability requires clearer accounts of how they participate in bureaucratic judgment, how responsibility is distributed, and how affected people can challenge AI-mediated decisions. Therefore, future inventories should move toward documenting the practices through which AI systems are classified, reviewed, monitored, and contested.

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